

SON LA, Viet Nam

WMO: 488060

Lat: 21.3322N

Lon: 103.9060E

Elev: 2218

StdP: 13.56

Time Zone: 7.00 (E07)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.7	47.3	36.4	34.2	53.5	39.9	39.3	54.2	15.7	57.9	13.5	60.2	2.9	140	0.253

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.8	91.8	72.2	89.9	73.1	88.3	73.8	78.0	85.7	77.3	84.8	76.7	83.9	6.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.0	147.9	81.6	75.4	144.4	80.7	74.7	141.3	79.9	43.5	86.2	42.8	85.2	42.1	84.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	11.2	9.5	39.8	94.9	3.2	1.7	37.5	96.1	35.6	97.0	33.8	98.0	31.5	99.2		
(5)			38.5	80.9	2.8	2.0	36.4	82.4	34.8	83.6	33.2	84.7	31.2	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.1	60.7	64.7	70.6	75.8	78.0	79.1	78.6	78.3	76.8	73.4	67.6	61.8		
	DBStd	8.06	6.08	7.50	6.26	5.15	4.04	2.69	2.33	2.20	2.74	3.47	4.88	5.43		
	HDD50	8	3	4	0	0	0	0	0	0	0	0	0	1		
	HDD65	431	155	89	28	2	0	0	0	0	0	1	29	127		
	CDD50	8090	336	416	638	774	867	872	885	878	804	727	527	366		
	CDD65	3037	22	81	201	326	402	422	420	413	354	263	106	27		
	CDH74	22153	193	654	1530	2734	3275	3381	3031	2896	2347	1425	551	136		
	CDH80	6714	19	166	559	1126	1216	1015	825	798	627	283	73	7		
(14) Wind	WSAvg	3.3	3.7	4.1	4.2	3.8	3.4	3.1	2.8	2.8	2.5	2.7	2.9	3.3		
(15) Precipitation	PrecAvg	55.0	1.0	0.9	2.1	4.2	7.8	8.4	10.7	10.1	5.1	2.2	1.4	1.1		
	PrecMax	79.8	4.1	3.0	5.3	10.9	15.0	14.8	19.3	16.5	16.6	6.7	5.6	4.4		
	PrecMin	40.8	0.0	0.0	0.0	1.6	1.9	3.3	5.3	4.3	1.6	0.0	0.0	0.0		
	PrecStd	9.1	1.2	0.9	1.3	1.9	3.3	3.1	3.4	3.6	3.0	1.6	1.5	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.5	87.3	92.0	94.7	94.5	91.7	90.5	89.9	89.3	87.1	84.5	80.9		
		MCWB	62.3	64.5	65.6	69.7	72.6	75.6	77.3	76.9	75.8	73.4	70.4	66.9		
	2%	DB	79.0	84.7	89.2	91.8	91.7	88.9	88.2	87.9	87.4	84.8	82.2	77.6		
		MCWB	63.4	64.0	66.3	71.2	73.8	76.1	76.6	76.4	75.1	72.6	69.3	65.2		
	5%	DB	75.7	82.0	86.2	89.6	89.2	87.2	86.4	86.3	85.7	83.3	80.1	74.9		
		MCWB	63.1	64.7	67.2	71.8	74.2	75.8	76.4	76.1	74.6	71.9	68.7	64.4		
	10%	DB	71.9	78.3	82.9	86.9	86.7	85.5	84.6	84.5	84.0	81.4	77.4	71.6		
		MCWB	62.1	64.8	67.9	72.0	74.0	75.6	76.0	75.8	73.8	71.2	67.0	62.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.8	70.5	72.9	76.6	77.8	78.8	79.1	78.7	77.7	75.9	73.5	69.4		
		MCDB	77.4	81.3	84.2	87.7	87.0	87.0	86.6	86.1	85.7	83.5	80.9	75.8		
	2%	WB	65.6	68.3	71.2	75.1	76.6	77.6	78.0	77.5	76.5	74.3	71.7	67.6		
		MCDB	73.9	78.8	81.8	85.7	85.5	85.1	85.2	84.9	84.1	82.0	78.7	74.6		
	5%	WB	64.0	66.9	70.0	74.0	75.8	76.9	77.2	76.8	75.7	73.2	70.1	65.4		
		MCDB	72.1	76.5	80.2	84.3	84.4	84.1	84.0	84.1	82.7	80.6	76.6	72.5		
	10%	WB	62.5	65.4	68.8	73.0	75.0	76.2	76.4	76.1	74.9	72.2	68.7	63.7		
		MCDB	70.1	74.3	78.1	82.8	83.2	82.8	82.6	82.6	81.3	78.8	74.9	70.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	18.0	18.2	17.0	14.6	11.8	11.1	11.8	13.5	14.2	16.3	16.2		
		MCDBR	24.2	26.6	25.8	21.6	18.6	14.8	14.5	14.9	16.4	17.1	19.6	21.1		
	5% WB	MCWBR	10.3	9.4	8.4	6.7	5.4	4.7	5.2	5.4	5.7	6.2	8.1	9.6		
		MCWBR	19.8	21.9	20.5	19.0	15.1	12.8	12.6	13.0	13.8	14.7	17.0	18.3		
(40) Clear-Sky Solar Irradiance	taub		0.475	0.538	0.761	0.741	0.488	0.447	0.444	0.487	0.517	0.563	0.485	0.498		
	taud		2.011	1.871	1.444	1.535	2.188	2.332	2.328	2.182	2.067	1.885	2.039	1.947		
	Ebn at Noon		248	240	195	203	260	268	269	259	250	230	244	235		
	Edh at Noon		51	62	100	92	48	41	41	48	53	61	49	53		
(44) All-Sky Solar Radiation	RadAvg		940	1213	1332	1547	1552	1390	1391	1415	1391	1185	1141	939		
	RadStd		135	193	117	157	144	117	123	102	74	92	119	107		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.03	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+0.63	N/A	-1.40	+1.12	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page